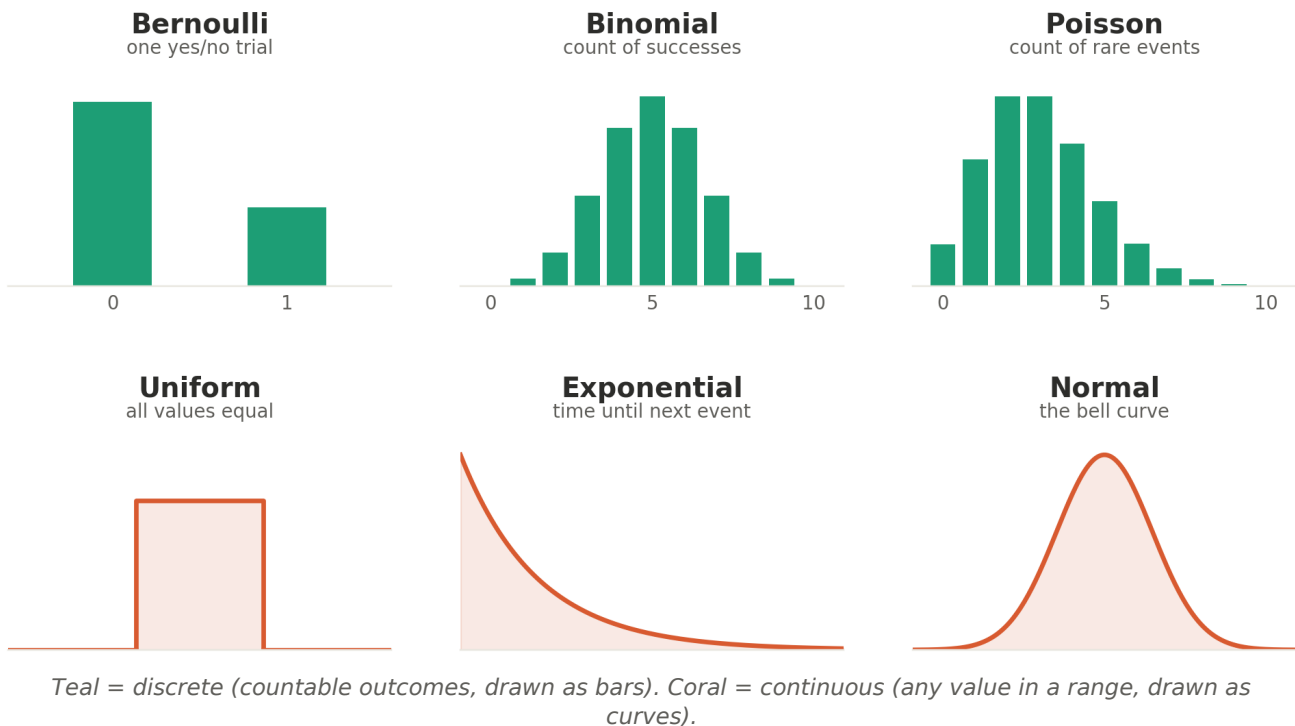


Six probability distributions

A trainer's reference: what each one models, how to tell them apart, and how they connect. The one question that sorts all six: are you **counting** something (discrete) or **measuring** something on a continuous scale (continuous)?



The discrete trio: you are counting

Bernoulli. The atom. One trial, two outcomes: success or failure, 1 or 0. The only knob is p , the probability of success. Everything below is built from this. On average you get p .

e.g. a single coin flip ($p = 0.5$); one unit passing or failing inspection ($p = \text{pass rate}$).

Binomial. Bernoulli repeated n independent times, counting how many successes. Knobs: n and p . The shape piles up around the average $n \times p$ and looks more bell-shaped as n grows. Key signal: a fixed number of yes/no trials, so there is a denominator ("x out of n").

e.g. heads in 10 flips; defective units in a batch of 100; how many of 50 customers said yes.

Poisson. Also counting, but with no fixed number of trials. You count events in a window of time, space, or area, where events are rare, independent, and arrive at a steady average rate λ (lambda). There is no denominator. A handy fingerprint: mean and variance are both λ , so if count data has mean $\approx$ variance, suspect Poisson. Right-skewed for small λ , straightening toward a bell as λ grows.

e.g. machine breakdowns per week; customer arrivals per hour; defects per square metre; typos per page.

Quick discriminator: Binomial has a denominator ("12 out of 100"); Poisson does not ("12 events this week").

The continuous trio: you are measuring

Uniform. The flat one. Every value in a range $[a, b]$ is equally likely, so the shape is a flat plateau. It is the "I know the range and nothing more" distribution: maximum ignorance about what is inside. The mean is just the midpoint, $(a + b) / 2$.

e.g. a random number between 0 and 1; rounding error to the nearest mm; where a single flaw sits on a wire.

Exponential. The time you wait until the next Poisson event. This is the crucial pairing: Poisson counts the events, Exponential measures the gaps between them (same knob λ , the rate). The shape decays from a peak at zero: short waits are common, long waits rapidly rarer. Its quirk is being memoryless: a part that has run 100 hours has the same remaining-life distribution as a new one. That is why it models random failure but not wear-out (wear-out needs a Weibull).

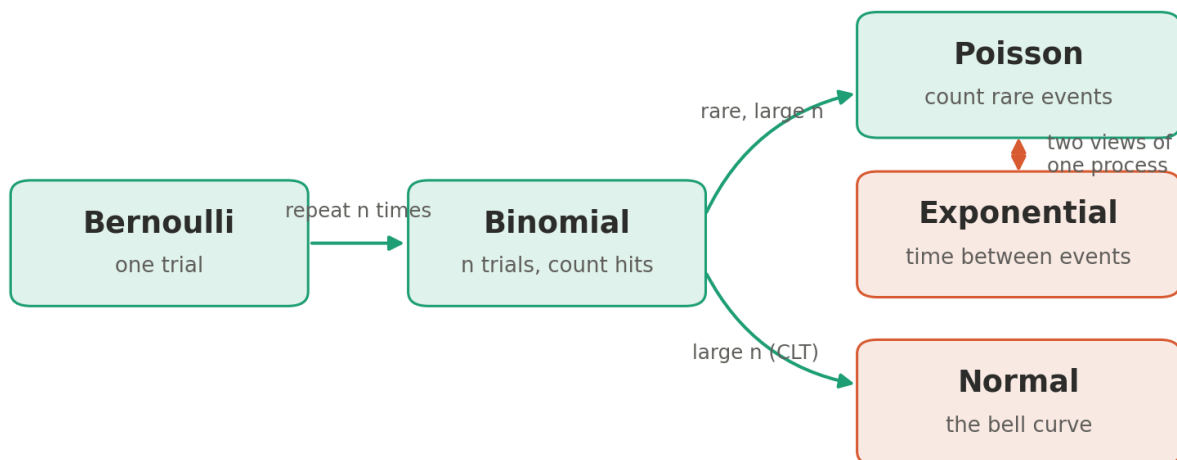
e.g. time until the next machine failure; time between customer arrivals.

Normal. The bell, and it is everywhere because of the Central Limit Theorem: whenever an outcome is the sum or average of many small independent effects, it tends toward Normal regardless of the pieces. Knobs: μ (centre) and σ (spread). The 68-95-99.7 rule: about 68% of values fall within 1σ of the mean, 95% within 2σ , 99.7% within 3σ . This is the foundation of SPC, Cp/Cpk, and the "six sigma" name itself.

e.g. a machined part's diameter; package fill weight; adult heights.

How they connect

They are not six unrelated objects but a family with bloodlines. Stack n Bernoulli trials and you get Binomial. Push Binomial to many trials with a rare event (large n , tiny p) and it converges to Poisson. Push it to many trials generally and the Central Limit Theorem bends it into a Normal. Poisson and Exponential are the same process seen two ways: one counts the events, the other times the gaps.



Quick reference

Distribution	Reach for it when you are...	Knobs	Everyday example
Bernoulli	scoring one yes/no trial	p	a single coin flip
Binomial	counting hits in n fixed trials	n, p	defects in a batch of 100
Poisson	counting rare events per interval	λ	breakdowns per week
Uniform	only knowing the range	a, b	a random number 0 to 1
Exponential	timing the wait to the next event	λ	time to next failure
Normal	measuring a sum/average of effects	μ, σ	a machined part's diameter

How to choose, in one breath

Counting or measuring? If counting: one trial (Bernoulli), a fixed batch (Binomial), or open-ended rare events (Poisson)? If measuring: flat across a range (Uniform), a waiting time (Exponential), or a centred bell (Normal)?